Depth Pseudo Color Image

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1. Course Content

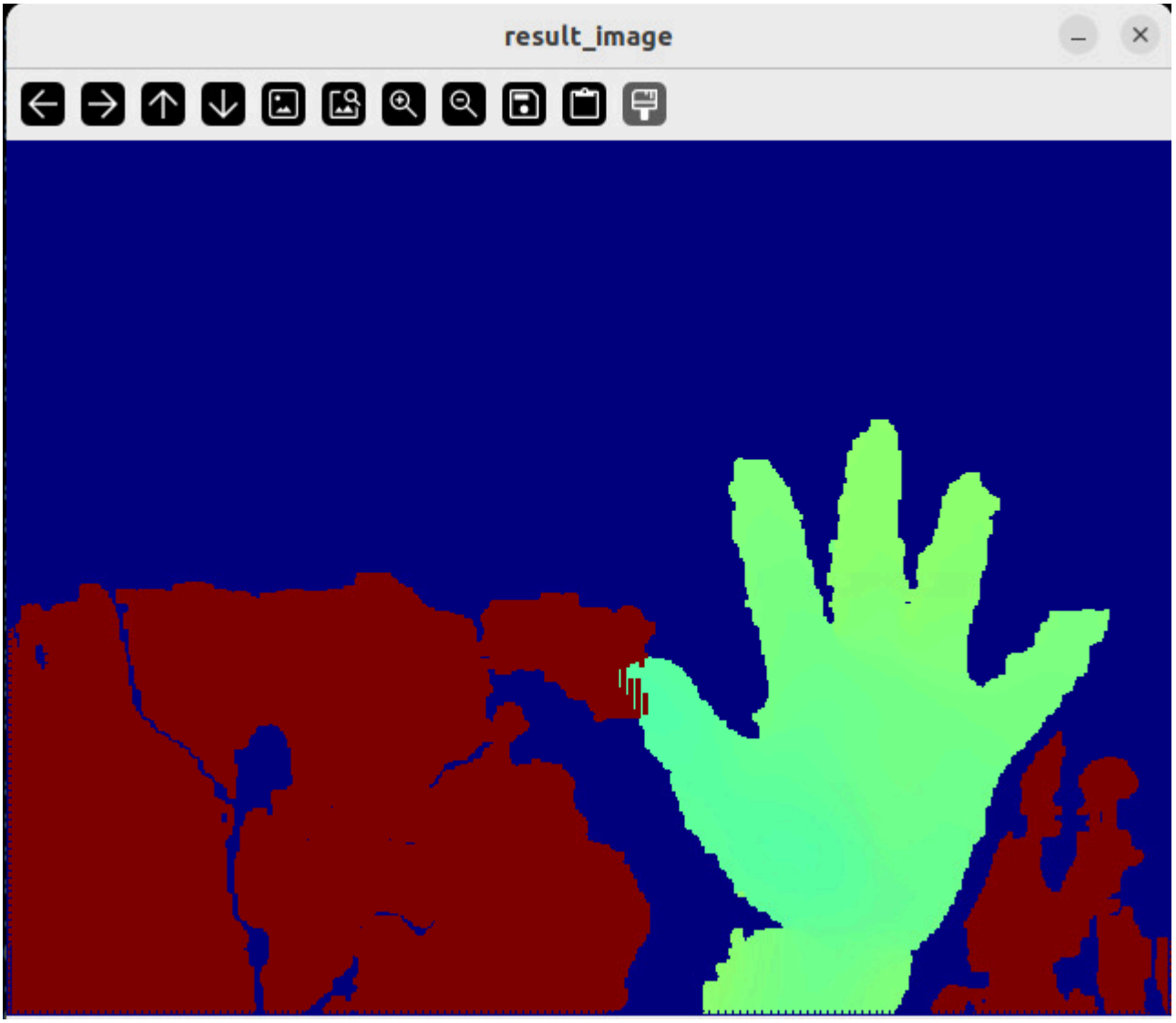
Run the example program; by subscribing to the depth color image topic and using opencv image processing, convert the black-and-white depth image into a pseudo color image.

> **Important**
>
> This section is for reference only for the depth camera.

2. Case Demonstration

```
ros2 launch yahboomcar_bringup camera_demo.launch.py demo:=GetDepthColor
```

After startup, as shown in the figure below:



Different colors indicate differences in depth information.

3. Source Code Analysis

- Source code path:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/common_demo/common_demo/camera/GetDepthColor.py
```

- Import the required library files

```
from cv_bridge import CvBridge
import cv2
from rclpy.node import Node
import rclpy
from sensor_msgs.msg import Image
```

- Define the depth image decoding format

```
encoding = ['16UC1', '32FC1']
```

- Define the subscriber to subscribe to the depth image topic

```
self.sub_depth =
self.create_subscription(Image,"/camera/depth/image_raw",self.get_DepthImgCallBack,100)
```

- Define `self.depth_bridge` to convert the message format into an image format that opencv can process

```
self.depth_bridge = CvBridge()
```

- Convert the image data message into an image

```
depth_image = self.depth_bridge.imgmsg_to_cv2(depth_frame, encoding[1])
```

- Call the opencv image processing function `cv2.applyColorMap` to convert the depth image

```
depth_to_color_image = cv2.applyColorMap(cv2.convertScaleAbs(depth_image, alpha=1.0),
cv2.COLORMAP_JET)
```

cv2.convertScaleAbs: performs a **linear transformation** on the image data and converts it to the **8-bit unsigned integer (uint8)** format, with scaling factor as the scaling factor.

cv2.applyColorMap: a function that converts a **single-channel grayscale image** into a **pseudo-color image**, enhancing the visualization effect through color mapping. This parameter has two variables: the first is a single-channel 8-bit or floating-point image (grayscale image), and the second is the color map, which can be one of the following values,

```
cv2.COLORMAP_AUTUMN         #Red-Orange-Yellow
cv2.COLORMAP_BONE           #Black-White (grayscale with a bluish tint)
cv2.COLORMAP_HSV            #HSV color space
cv2.COLORMAP_JET            #Blue-Cyan-Yellow-Red (classic heatmap)
cv2.COLORMAP_WINTER         #Blue-Green
cv2.COLORMAP_RAINBOW        #Rainbow colors (Red-Purple)
cv2.COLORMAP_OCEAN          #Dark blue-Light blue
cv2.COLORMAP_SUMMER         #Green-Yellow
cv2.COLORMAP_SPRING         #Magenta-Yellow
cv2.COLORMAP_COOL           #Cyan-Magenta
cv2.COLORMAP_HSV            #HSV color space loop
cv2.COLORMAP_PINK           #Pink tones
cv2.COLORMAP_PARULA         #MATLAB Parula style
cv2.COLORMAP_VIRIDIS        #Modern scientific color scale (Green-Purple)
cv2.COLORMAP_INFERNO        #Black-Red-Yellow-White (high contrast)
cv2.COLORMAP_PLASMA         #Purple-Red-Yellow
cv2.COLORMAP_MAGMA          #Black-Red-White
cv2.COLORMAP_TWILIGHT       #Blue-Pink-White gradient
```